

CS 736: Medical Image Computing

(Instructor : Suyash P. Awate)

Mid-Sem Exam: Closed Book / Notes / Everything

Roll Number: _____

Name: _____

When in doubt, make suitable assumptions, state them clearly, and proceed to solve the problem. All the best!

Formulae for Reference

Radon Transform: The Radon transform of a function $f(x, y)$ is a function $Rf(t, \theta)$ where for any given t, θ , the value of the Radon transform is

$$Rf(t, \theta) := \int_{s=-\infty}^{\infty} f(x(s), y(s)) ds,$$

where $x(s) := t \cos \theta - s \sin \theta, y(s) := t \sin \theta + s \cos \theta$

Fourier Transform: The 1D Fourier transform of an integrable function $f(x)$ is defined for each real number w as:

$$F_1 f(w) := \int_{x=-\infty}^{\infty} f(x) e^{-iwx} dx$$

Inverse Fourier Transform: For an integrable function $h(w)$, the 1D inverse Fourier transform is defined for each real number x as:

$$F_1^{-1} h(x) := \frac{1}{2\pi} \int_{w=-\infty}^{\infty} h(w) e^{iwx} dw$$

Central Slice Theorem: For a 2D function $f(x, y)$, the values of the (i) 2D Fourier transform along a line through origin making an angle θ with the $+X$ axis are the same as the values of the (ii) 1D Fourier transform of the Radon transform along angle θ , i.e.,

$$F_2 f(w \cos \theta, w \sin \theta) = F_1 (Rf(\cdot, \theta))(w),$$

where $F_2 f(\cdot, \cdot)$ is the 2D Fourier transform of $f(\cdot, \cdot)$.

1. [20 points]

Consider an image reconstruction problem on a spatio-temporal computed-tomography (CT) image X , i.e., a temporal sequence of 2D CT images.

Suppose you have prior information that the spatial variation of intensities in X , at any time point t , is smooth (without edges) and the temporal variation of intensities, at any pixel q , is piecewise smooth.

Let Y represent the spatiotemporal data acquired (measurements) by the CT scanner.

- (5 points) Design a suitable likelihood model $P(Y|X)$.

Negative log likelihood $\propto \|Y^t - A^t X^t\|_2^2$

- (5 points) Design a suitable prior model $P(X)$.

Negative log prior $\propto \sum_t \sum_i \sum_{j \in \mathcal{N}_i} \alpha (X_j^t - X_i^t)^2 + \sum_i \sum_t \beta H_\delta(X_i^{t+1} - X_i^t)$
where $H_\delta(\cdot)$ is the Huber loss function.

- (5 points) Design a Bayesian optimization problem for reconstructing the spatiotemporal image X from the acquired data Y .
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Follows from the above.

- (5 points) Write down a step-by-step algorithm for solving the optimization problem, where every step is precisely defined in terms of mathematical and coding aspects (beyond simply a natural-language description).
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Please see lecture notes.

2. [10 points]

Consider a 1D $f(x)$ on a discrete domain $x = -20, -19, \dots, 19, 20$, where the real-valued image pixel intensities $f(x)$ are missing for the pixels at coordinates $x = 1, \dots, 9$. It is known that $f(0) = 0$ and $f(10) = 1$. Consider the problem of estimating the missing intensities $f(1), \dots, f(9)$ formulated as an optimization problem as follows:

$$\arg \min_{f(1), \dots, f(9)} \sum_{x=1}^{10} h(f(x) - f(x-1)),$$

where the function $h(\cdot)$ is the regularizing function.

- (5 points) If the function $h(\cdot)$ is defined as $h(a) := a^2$, then give the optimal solution(s) to the problem. Give a mathematical proof of the optimality of the solution. Is there a unique optimum or are there multiple optima? If there are multiple optima, describe them all.

Take the gradient of the objective function. Assign it to zero. We will get 9 equations with 9 unknowns. These can be solved sequentially to get $f(i) = i/10$ for $i = 1, 2, \dots, 9$.

- (5 points) If the function $h(\cdot)$ is defined as $h(a) := |a|$, then give the optimal solution(s) to the problem. Give a proof or a logical argument for the optimality of the solution. Is there a unique optimum or are there multiple optima ? If there are multiple optima, describe them all.
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It can be argued that the function $f(x)$ must be monotonically increasing. If it is not, then we can construct a suitable monotonic function that has a lower energy (proof by contradiction). Then, we see that there are many possible solutions, all producing the same value of the objective function, i.e., 1.

3. [15 points]

Consider a microscopy system that introduces two kinds of measurement errors in the grayscale images that it produces: (i) at some pixels, it leads to a NaN measurement that clearly needs to be ignored (or considered as missing data) and (ii) at all other pixels it introduces i.i.d. additive zero-mean Gaussian noise to the true incoming signal. The variance of the Gaussian noise is provided by the manufacturer at σ^2 .

You want to restore the missing pixel values as well as reduce the noise and the blur.

- (5 + 5 points) Formulate this problem as a MAP optimization problem (give all details for the likelihood and prior models).

Likelihood model can be Gaussian imposed only on those pixels for which data is available.

Prior model can be a MRF prior imposing piecewise-smooth a model OR a dictionary model (you can assume that a dictionary is given to you).

- (5 points) Write down a step-by-step algorithm for optimization, where every step is precisely defined in terms of mathematical and coding aspects (beyond simply a natural-language description).
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Please see lecture notes.

4. [10 points]

- Describe the algebraic reconstruction technique (ART) for CT reconstruction. Write down a step-by-step algorithm where every step is precisely defined in terms of mathematical and coding aspects (beyond simply a natural-language description).

See the lecture notes.

5. [15 points]

Describe the inverse Radon transform as follows.

- (5 points) Describe the backprojection operator mathematically analytically (not using matrices), as mapping between data in the (t, θ) domain to the image values in the spatial (x, y) domain.

See the lecture notes.

- (5 points) Describe the Ram-Lak filter and the Cosine filter.

See the lecture notes.

- (5 points) Describe the filtered-backprojection operator mathematically analytically (not using matrices), as mapping between data in the (t, θ) domain to the image values in the spatial (x, y) domain.
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See the lecture notes.